G2 Differentiating functions

Name:

Class:

Date:

Time: **464 minutes**

Marks: **383 marks**

Comments:

Q1.

(a) Find $\frac{dy}{dx}$ when:

(i) $y = (4x^2 + 3x + 2)^{10}$;

(2)

(ii) $y = x^2 \tan x$.

(2)

(b) (i) Find $\frac{dx}{dy}$ when $x = 2y^3 + \ln y$.

(1)

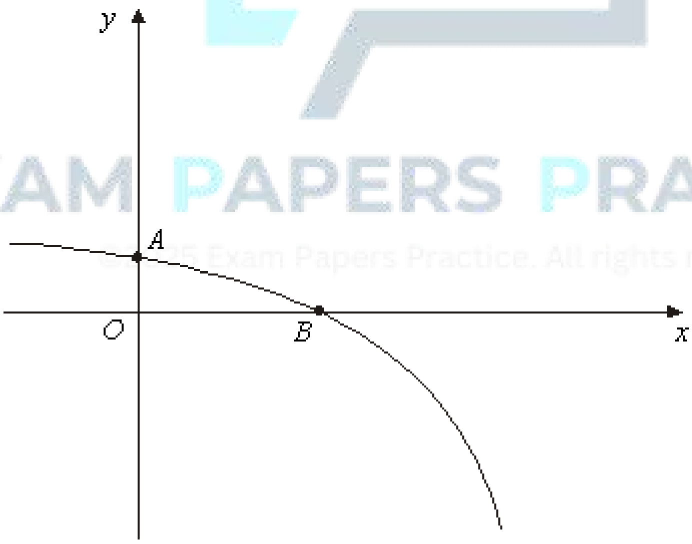
(ii) Hence find an equation of the tangent to the curve $x = 2y^3 + \ln y$ at the point (2,1).

(3)

(Total 8 marks)

Q2.

The sketch shows the graph of $y = 4 - e^{2x}$. The curve crosses the y-axis at the point A and the x-axis at the point B.



(a) (i) Find $\int (4 - e^{2x})\,dx$.

(2)

(ii) Hence show that $\int_0^{\ln 2} (4 - e^{2x})\,dx = 4\ln 2 - \frac{3}{2}$.

(2)

(b) (i) Write down the y-coordinate of A.

(1)

(ii) Show that $x = \ln 2$ at B.

(2)

(c) Find the equation of the normal to the curve $y = 4 - e^{2x}$ at the point B.

(4)

(d) Find the area of the region enclosed by the curve $y = 4 - e^{2x}$, the normal to the curve at B and the y-axis.

(3)

(Total 14 marks)

Q3.

(a) (i) Solve the differential equation $\frac{dy}{dt} = y \sin t$ to obtain y in terms of t.

(4)

(ii) Given that $y = 50$ when $t = \pi$, show that $y = 50e^{-(1 + \cos t)}$.

(3)

(b) A wave machine at a leisure pool produces waves. The height of the water, y cm, above a fixed point at time t seconds is given by the differential equation

$$\frac{dy}{dt} = y \sin t$$

(i) Given that this height is 50 cm after π seconds, find, to the nearest centimetre, the heigh tof the water after 6 seconds.

(2)

(ii) Find $\frac{d^2 y}{dt^2}$ and hence verify that the water reaches a maximum height after π seconds.

(4)

(Total 13 marks)

Q4.

A model helicopter takes off from a point O at time $t = 0$ and moves vertically so that its height, y cm, above O after time t seconds is given by

$$y = \frac{1}{4}t^4 - 26t^2 + 96t, \qquad 0 \le t \le 4$$

(a) Find:

(i) $\dfrac{dy}{dt}$;

(3)

(ii) $\dfrac{d^2y}{dt^2}$

(2)

(b) Verify that y has a stationary value when t = 2 and determine whether this stationary value is a maximum value or a minimum value.

(4)

(c) Find the rate of change of y with respect to t when t = 1

(2)

(d) Determine whether the height of the helicopter above O is increasing or decreasing at the instant when t = 3.

(2)
(Total 13 marks)

Q5.

A curve is defined for x > 0 by the equation

$$y=\left(1+\frac{2}{x}\right)^2$$

The point P lies on the curve where x = 2.

(a) Find the y-coordinate of P.

(1)

(b) Expand $\left(1+\dfrac{2}{x}\right)^2$.

(2)

(c) Find $\dfrac{dy}{dx}$

(3)

(d) Hence show that the gradient of the curve at P is – 2.

(2)

(e) Find the equation of the normal to the curve at P, giving your answer in the form $x + by + c = 0$, where b and c are integers.

(4)
(Total 12 marks)

Q6.

(a) Differentiate ln x with respect to x.

(1)

(b) Given that $y = (x + 1) \ln x$, find $\frac{dy}{dx}$.

(2)

(c) Find an equation of the normal to the curve $y = (x + 1) \ln x$ at the point where $x = 1$.

(4)
(Total 7 marks)

Q7.

A biologist is researching the growth of a certain species of hamster. She proposes that the length, x cm, of a hamster t days after its birth is given by

$$x = 15 - 12e^{-\frac{t}{14}}$$

(a) Use this model to find:

(i) the length of a hamster when it is born;

(1)

(ii) the length of a hamster after 14 days, giving your answer to three significant figures.

(2)

(b) (i) Show that the time for a hamster to grow to 10 cm in length is given by $t = 14 \ln\left(\frac{a}{b}\right)$, where a and b are integers.

(3)

(ii) Find this time to the nearest day.

(1)

(c) (i) Show that

$$\frac{dx}{dt} = \frac{1}{14}(15 - x)$$

(3)

(ii) Find the rate of growth of the hamster, in cm per day, when its length is 8 cm.

(1)
(Total 11 marks)

Q8.

The point $P(1, a)$, where $a > 0$, lies on the curve $y + 4x = 5x^2y^2$.

(a) Show that $a = 1$.

(2)

(b) Find the gradient of the curve at P.

(7)

(c) Find an equation of the tangent to the curve at P.

(1)
(Total 10 marks)

Q9.

The volume, V m³, of water in a tank at time t seconds is given by

$$V = \frac{1}{3}t^6 - 2t^4 + 3t^2, \quad \text{for } t \geq 0$$

(a) Find:

(i) $\frac{dV}{dt}$;

(3)

(ii) $\frac{d^2V}{dt^2}$.

(2)

(b) Find the rate of change of the volume of water in the tank, in $m^3 s^{-1}$, when $t = 2$.

(2)

(c) (i) Verify that V has a stationary value when $t = 1$.

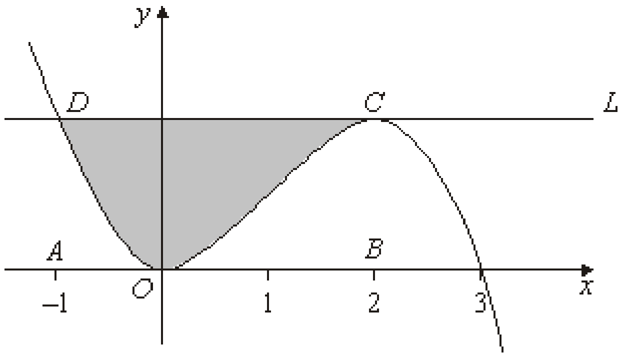
(2)

(ii) Determine whether this is a maximum or minimum value.

(2)
(Total 11 marks)

Q10.

The diagram shows the curve with equation $y = 3x^2 - x^3$ and the line L.



The points A and B have coordinates (–1, 0) and (2, 0) respectively. The curve touches the x-axis at the origin O and crosses the x-axis at the point (3, 0). The line L cuts the curve at the point D where $x = -1$ and touches the curve at C where $x = 2$.

(a) Find the area of the rectangle $ABCD$.

(2)

(b) (i) Find $\int (3x^2 - x^3)\,dx$.

(3)

(ii) Hence find the area of the shaded region bounded by the curve and the line L.

(4)

(c) For the curve above with equation $y = 3x^2 - x^3$:

(i) find $\frac{dy}{dx}$;

(2)

(ii) hence find an equation of the tangent at the point on the curve where $x = 1$;

(3)

(iii) show that y is decreasing when $x^2 - 2x > 0$.

(2)

(d) Solve the inequality $x^2 - 2x > 0$.

(2)

(Total 18 marks)

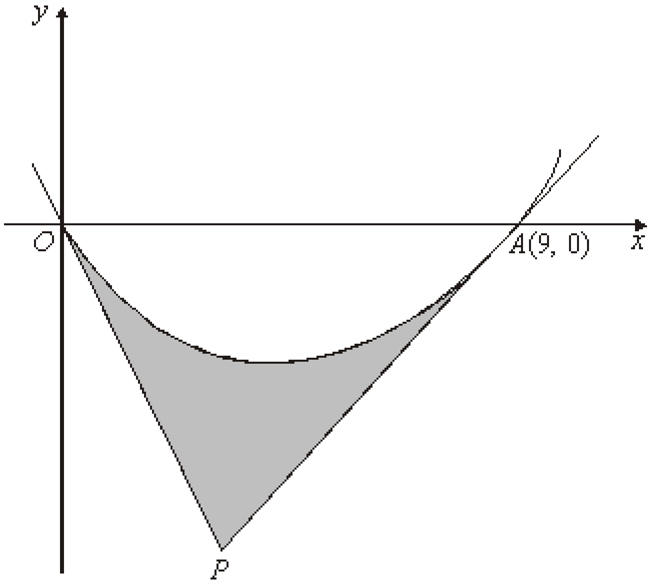
Q11.

Given that $y = 16x + x^{-1}$, find the two values of x for which $\frac{dy}{dx} = 0$.

(Total 5 marks)

Q12.

A curve, drawn from the origin O, crosses the x-axis at the point $A(9, 0)$. Tangents to the curve at O and A meet at the point P, as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = x^{\frac{3}{2}} - 3x$$

(a) Find $\frac{dy}{dx}$.

(2)

(b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O.

(2)

(ii) Show that the equation of the tangent at $A(9, 0)$ is $2y = 3x - 27$.

(3)

(iii) Hence find the coordinates of the point P where the two tangents meet.

(3)

(c) Find $\int \left(x^{\frac{3}{2}} - 3x \right) dx$.

(3)

(d) Calculate the area of the shaded region bounded by the curve and the tangents OP and AP.

(5)

(Total 18 marks)

Q13.

(a) Find $\frac{dy}{dx}$ when $y = \tan 3x$.

(2)

(b) Given that $y = \frac{3x+1}{2x+1}$, show that $\frac{dy}{dx} = \frac{1}{(2x+1)^2}$.

(3)
(Total 5 marks)

Q14.

(a) (i) Given that $f(x) = x^4 + 2x$, find $f'(x)$.

(1)

(ii) Hence, or otherwise, find $\int \frac{2x^3+1}{x^4+2x}\,dx$.

(2)

(b) (i) Use the substitution $u = 2x + 1$ to show that

$$\int x\sqrt{2x+1}\,dx = \frac{1}{4}\int\left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right)du$$

(3)

(ii) Hence show that $\int_0^4 x\sqrt{2x+1}\,dx = 19.9$ correct to three significant figures.

(4)
(Total 10 marks)

Q15.

A curve has equation $y = 7 - 2x^5$.

(a) Find $\frac{dy}{dx}$.

(2)

(b) Find an equation for the tangent to the curve at the point where $x = 1$.

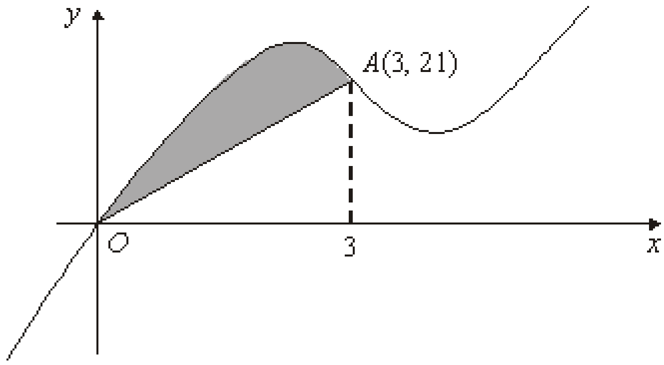
(3)

(c) Determine whether y is increasing or decreasing when $x = -2$.

(2)
(Total 7 marks)

Q16.

The curve with equation $y = x^3 - 10x^2 + 28x$ is sketched below.



The curve crosses the x-axis at the origin O and the point $A(3, 21)$ lies on the curve.

(a) (i) Find $\frac{dy}{dx}$.

(3)

(ii) Hence show that the curve has a stationary point when $x = 2$ and find the x-coordinate of the other stationary point.

(4)

(b) (i) Find $\int (x^3 - 10x^2 + 28x)\,dx$.

(3)

(ii) Hence show that $\int_0^3 (x^3 - 10x^2 + 28x)\,dx = 56\frac{1}{4}$.

(2)

(iii) Hence determine the area of the shaded region bounded by the curve and the line OA.

(3)
(Total 15 marks)

Q17.

At the point (x, y), where $x > 0$, the gradient of a curve is given by

$$\frac{dy}{dx} = 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7$$

(a) (i) Verify that $\frac{dy}{dx} = 0$ when $x = 4$.

(1)

(ii) Write $\frac{16}{x^2}$ in the form $16x^k$, where k is an integer.

(1)

(iii) Find $\frac{d^2y}{dx^2}$.

(3)

(iv) Hence determine whether the point where $x = 4$ is a maximum or a minimum, giving a reason for your answer.

(2)

(b) The point $P(1, 8)$ lies on the curve.

(i) Show that the gradient of the curve at the point P is 12.

(1)

(ii) Find an equation of the normal to the curve at P.

(3)

(c) (i) Find $\int (3x^{\frac{1}{2}} + \frac{16}{x^2} - 7)dx$

(3)

(ii) Hence find the equation of the curve which passes through the point P(l, 8).

(3)

(Total 17 marks)

Q18.

(a) A curve has equation $y = e^{2x} - 10e^x + 12x$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Find $\frac{d^2y}{dx^2}$.

(1)

(b) The points P and Q are the stationary points of the curve.

(i) Show that the x-coordinates of P and Q are given by the solutions of the equation

$$e^{2x} - 5e^x + 6 = 0$$

(1)

(ii) By using the substitution $z = e^x$, or otherwise, show that the x-coordinates of P and Q are $\ln 2$ and $\ln 3$.

(3)

(iii) Find the y-coordinates of P and Q, giving each of your answers in the form $m + 12 \ln n$, where m and n are integers.

(3)

(iv) Using the answer to part (a)(ii), determine the nature of each stationary point.

(3)

(Total 13 marks)

Q19.

(a) Given that $y = x \ln x$, find $\frac{dy}{dx}$.

(2)

(b) Hence, or otherwise, find $\int \ln x \, dx$.

(2)

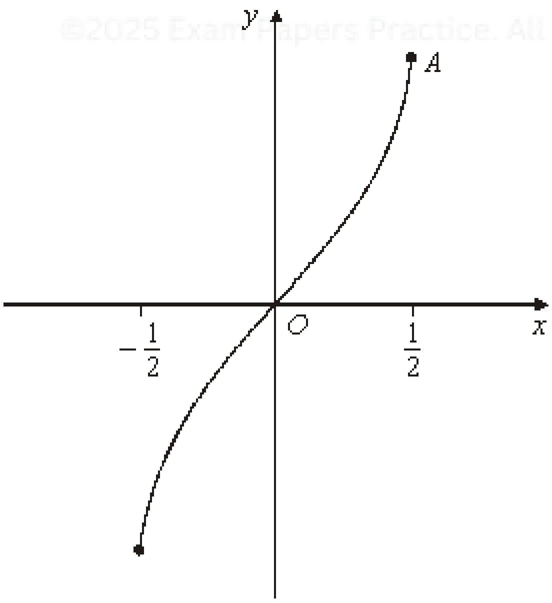
(c) Find the exact value of $\int_1^5 \ln x \, dx$.

(2)

(Total 6 marks)

Q20.

The diagram shows the curve with equation $y = \sin^{-1} 2x$, where $-\frac{1}{2} \leq x \leq \frac{1}{2}$.



(a) Find the y-coordinate of the point A, where $x = \frac{1}{2}$.

(1)

(b) (i) Given that $y = \sin^{-1} 2x$, show that $x = \frac{1}{2} \sin y$.

(1)

(ii) Given that $x = \frac{1}{2} \sin y$, find $\frac{dx}{dy}$ in terms of y.

(1)

(c) Using the answers to part (b) and a suitable trigonometrical identity, show that

$$\frac{dx}{dy} = \frac{2}{\sqrt{1-4x^2}}$$

(4)

(Total 7 marks)

Q21.

A curve is defined by the equation

$y^2 - xy + 3x^2 - 5 = 0$

(a) Find the y-coordinates of the two points on the curve where $x = 1$.

(3)

(b) (i) Show that $\frac{dy}{dx} = \frac{y-6x}{2y-x}$.

(6)

(ii) Find the gradient of the curve at each of the points where $x = 1$.

(2)

(iii) Show that, at the two stationary points on the curve, $33x^2 - 5 = 0$.

(3)

(Total 14 marks)

Q22.

A curve has equation $y = x^5 - 6x^3 - 3x + 25$.

(a) Find $\frac{dy}{dx}$.

(3)

(b) The point P on the curve has coordinates (2, 3).

(i) Show that the gradient of the curve at P is 5.

(2)

(ii) Hence find an equation of the normal to the curve at P, expressing your answer in the form $ax + by = c$, where a, b and c are integers.

(3)

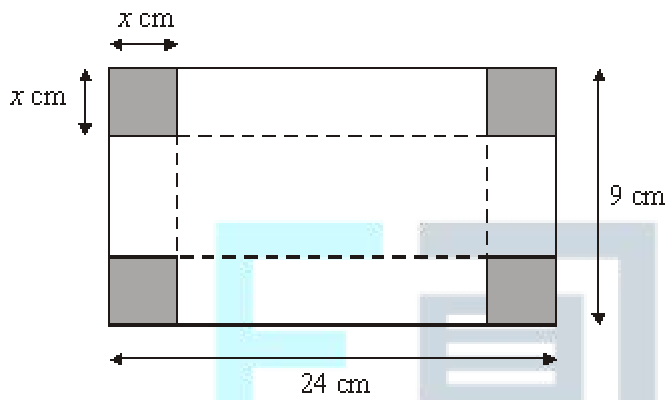
(c) Determine whether y is increasing or decreasing when $x = 1$.

(2)

(Total 10 marks)

Q23.

The diagram below shows a rectangular sheet of metal 24 cm by 9 cm.



A square of side x cm is cut from each corner and the metal is then folded along the broken lines to make an open box with a rectangular base and height x cm.

(a) Show that the volume, V cm^3, of liquid the box can hold is given by

$$V = 4x^3 - 66x^2 + 216x$$

(3)

(b) (i) Find $\frac{dV}{dx}$.

(3)

(ii) Show that any stationary values of V must occur when $x^2 - 11x + 18 = 0$

(2)

(iii) Solve the equation $x^2 - 11x + 18 = 0$.

(2)

(iv) Explain why there is only one value of x for which V is stationary.

(1)

(c) (i) Find $\frac{d^2V}{dx^2}$.

(2)

(ii) Hence determine whether the stationary value is a maximum or minimum.

(2)
(Total 15 marks)

Q24.

A curve is defined for $x > 0$ by the equation $y = x + \frac{2}{x}$.

(a) (i) Find $\frac{dy}{dx}$.

(3)

(ii) Hence show that the gradient of the curve at the point P where $x = 2$ is $\frac{1}{2}$.

(1)

(b) Find an equation of the normal to the curve at this point P.

(4)
(Total 8 marks)

Q25.

(a) (i) Using the binomial expansion, or otherwise, express $(2 + x)^3$ in the form $8 + ax + bx^2 + x^3$, where a and b are integers.

(3)

(ii) Write down the expansion of $(2 - x)^3$.

(2)

(b) Hence show that $(2 + x)^3 - (2 - x)^3 = 24x + 2x^3$.

(2)

(c) Hence show that the curve with equation

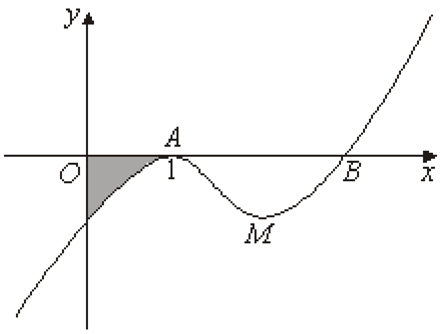
$$y = (2 + x)^3 - (2 - x)^3$$

has no stationary points.

(3)
(Total 10 marks)

Q26.

The curve with equation $y = x^3 - 5x^2 + 7x - 3$ is sketched below.



The curve touches the x-axis at the point A (1, 0) and cuts the x-axis at the point B.

(a) (i) Use the factor theorem to show that $x - 3$ is a factor of

$$p(x) = x^3 - 5x^2 + 7x - 3$$

(2)

(ii) Hence find the coordinates of B.

(1)

(b) The point M, shown on the diagram, is a minimum point of the curve with equation $y = x^3 - 5x^2 + 7x - 3$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Hence determine the x-coordinate of M.

(3)

(c) Find the value of $\frac{d^2y}{dx^2}$ when $x = 1$.

(2)

(d) (i) Find $\int (x^3 - 5x^2 + 7x - 3)\,dx$.

(4)

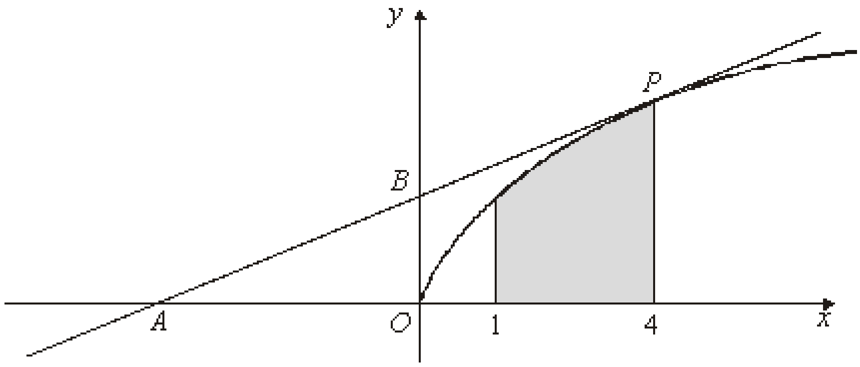
(ii) Hence determine the area of the shaded region bounded by the curve and the coordinate axes.

(4)

(Total 18 marks)

Q27.

The diagram shows a curve C with equation $y = \sqrt{x}$. The point O is the origin (0, 0).



The region bounded by the curve C, the x-axis and the vertical lines x = 1 and x = 4 is shown shaded in the diagram.

(a) (i) Write $\sqrt{x}$ in the form x^p, where p is a constant.

(1)

(ii) Find $\int \sqrt{x}\, dx$.

(2)

(iii) Hence find the area of the shaded region.

(3)

(b) The point on C for which x = 4 is P. The tangent to C at the point P intersects the x-axis and the y-axis at the points A and B respectively.

(i) Find an equation for the tangent to the curve C at the point P.

(4)

(ii) Find the area of the triangle AOB.

(3)

(c) Describe the single geometrical transformation by which the curve with equation $y = \sqrt{x-1}$ can be obtained from the curve C.

(2)

(d) Use the trapezium rule with four ordinates (three strips) to find an approximation for $\int_1^4 \sqrt{x-1}\, dx$, giving your answer to three significant figures.

(4)

(Total 19 marks)

Q28.

A curve is defined, for $x > 0$, by the equation y = f(x), where

$$f(x) = \frac{x^8 - 1}{x^3}$$

(a) Express $\frac{x^8 - 1}{x^3}$ in the form $x^p - x^q$, where p and q are integers.

(2)

(b) (i) Hence differentiate f(x) to find f′(x).

(2)

(ii) Hence show that f is an increasing function.

(2)

(c) Find the gradient of the normal to the curve at the point (1, 0).

(3)

(Total 9 marks)

Q29.

(a) Find $\int e^{4x} dx$.

(1)

(b) Use integration by parts to find $\int e^{4x}(2x + 1) dx$.

(3)

(c) By using the substitution $u = 1 + \ln x$, or otherwise, find $\int \frac{1 + \ln x}{x} dx$.

(4)

(Total 8 marks)

Q30.

A particle moves in a straight line. At time t seconds, it has velocity v m s^{-1}, where

$$v = 6t^2 - 2e^{-4t} + 8$$

and $t \geq 0$.

(a) (i) Find an expression for the acceleration of the particle at time t.

(2)

(ii) Find the acceleration of the particle when $t = 0.5$.

(2)

(b) The particle has mass 4 kg.

Find the magnitude of the force acting on the particle when $t = 0.5$.

(1)

(c) When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t.

(4)
(Total 9 marks)

Q31.

A particle moves in a horizontal plane under the action of a single force, **F** newtons. The unit vectors **i** and **j** are directed east and north respectively. At time t seconds, the velocity of the particle, **v** m s^{-1}, is given by

$$\mathbf{v} = 4e^{-2t}\,\mathbf{i} + (6t - 3t^2)\mathbf{j}$$

(a) Find an expression for the acceleration of the particle at time t.

(3)

(b) The mass of the particle is 5 kg.

(i) Find an expression for the force **F** acting on the particle at time t.

(2)

(ii) Find the magnitude of **F** when $t = 0$.

(2)

(c) Find the value of t when **F** acts due west.

(2)

(d) When $t = 0$, the particle is at the point with position vector (6**i** + 5**j**) m.

Find the position vector, **r** metres, of the particle at time t.

(5)
(Total 14 marks)

Q32.

A particle moves along a straight line through the origin. At time t, the displacement, s, of the particle from the origin is given by

$$s = 5t^2 + 3\cos 4t$$

Find the velocity of the particle at time t.

(Total 3 marks)

Q33.

A particle moves under the action of a force, **F** newtons. At time t seconds, the velocity, **v** m s^{-1}, of the particle is given by

$$\mathbf{v} = (t^3 - 15t - 5)\mathbf{i} + (6t - t^2)\mathbf{j}$$

(a) Find an expression for the acceleration of the particle at time t.

(3)

(b) The mass of the particle is 4 kg.

(i) Show that, at time t,

$$\mathbf{F} = (12t^2 - 60)\mathbf{i} + (24 - 8t)\mathbf{j}$$

(2)

(ii) Find the magnitude of **F** when $t = 2$.

(4)
(Total 9 marks)

Q34.

A particle moves in a straight line and at time t it has velocity v, where

$$v = 3t^2 - 2 \sin 3t + 6$$

(a) (i) Find an expression for the acceleration of the particle at time t.

(2)

(ii) When $t = \frac{\pi}{3}$, show that the acceleration of the particle is $2\pi + 6$.

(2)

(b) When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t.

(4)
(Total 8 marks)

Q35.

A particle moves in a horizontal plane under the action of a single force, **F** newtons. The unit vectors **i** and **j** are directed east and north respectively. At time t seconds, the position vector, **r** metres, of the particle is given by

$$\mathbf{r} = (t^3 - 3t^2 + 4)\mathbf{i} + (4t + t^2)\mathbf{j}$$

(a) Find an expression for the velocity of the particle at time t.

(2)

(b) The mass of the particle is 3 kg.

(i) Find an expression for **F** at time t.

(3)

(ii) Find the magnitude of **F** when $t = 3$.

(2)

(c) Find the value of t when **F** acts due north.

(2)

(Total 9 marks)